

Highlights

Using Data Categorization and Augmentation Strategies to improve Machine Learning Frameworks for Flame Spread over Electrical Wires

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- Non-linear ML models combined with smart augmentation methods push FSR prediction performance $R^2 > 0.9$.
- Adding categorical variables like insulation type and spread direction unlocks hidden patterns and sharply boosts model accuracy.
- Grouping data into physically meaningful regimes strengthens context-aware relationships and leads to more reliable flame-spread predictions.

Using Data Categorization and Augmentation Strategies to improve Machine Learning Frameworks for Flame Spread over Electrical Wires

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Abstract

Flame spread over electrical wires is influenced by a variety of factors including core and insulation materials, geometric dimensions, and environmental conditions. Traditional modeling approaches struggle to capture the complexity and non-linearity arising from these intertwined parameters. In this study, we combine machine learning (ML) algorithms with data categorization and augmentation strategies to improve predictive accuracy and interpretability of flame spread rate (FSR) models. We begin with a comprehensive dataset compiled from literature, encompassing a broad range of input parameters. A baseline is established using raw numerical data, which yields low to moderate predictive capability. Introducing categorical variables such as type of insulation and flame spread direction by using the One-Hot encoding technique improves the predictive capabilities of the models. This approach transformed classification features into binary indicators, allowing algorithms to leverage discrete categories more effectively unveiling hidden patterns, enhancing the model predictive capabilities. Non-linear models (e.g., Decision Trees, Gradient Boosting, Multi-Layer Perceptron) achieve R^2 values above 0.9. Data augmentation methods, including bootstrap resampling and Generative Adversarial Networks (GANs), further refine performance by filling data gaps and providing more robust training

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conditions.

The improved metrics—higher R^2 and lower MAE—reveal that clustering conditions into physically meaningful regimes guides the models to context-dependent relationships. The combination of categorization, augmentation, and cluster modeling yields more reliable predictions and deeper insights about input physical parameters and flame spread predictions. These findings suggest that integrating domain knowledge through categories, combined with advanced ML approaches, can significantly advance the theoretical modeling predictive capabilities for flame spread over electrical wires.

Keywords: Flame Spread Rate, Electrical Wires, Data Categorization, Supervised Learning.

1. Introduction

Electrical wires are potential sources of fires in buildings, industrial plants, and transportation (Babrauskas, 2016; Fernandez-Pello et al., 1991; Ahrens, 2016). However, they are also a source of fire hazards due to their susceptibility to ignition from excessive current, poor connections, and insulation breakdown (Kobayashi et al., 2021). Once ignited, flames can spread along wires, as polymer insulation and thermally conductive metal core can facilitate flame spread and potentially large-scale fire (Kobayashi et al., 2021; Bakhman et al., 1981). The complexity of flame spread in electrical wires arises from the interplay of gas and solid phase phenomena: the thermal conductivity and geometry of the wire core, the melting and dripping behavior of the insulation during burning, and environmental factors such as oxygen concentration, ambient pressure, gas flow velocity, and gravitational orientation (Kobayashi et al., 2021, 2018; Nakamura et al., 2008, 2009; Leung et al., 2000; Kikuchi et al., 1998; Zhao et al., 2017; Guo et al., 2024).

Investigations of flame spread over electrical wires have often employed simplified “laboratory wires”, composed of a metal core and a polymer tube insulation, to examine the effects of key parameters under controlled conditions (Kobayashi et al., 2021, 2018; Nakamura et al., 2008, 2009; Leung et al., 2000; Kikuchi et al., 1998; Zhao et al., 2017; Guo et al., 2024; Fujita et al., 2000, 2002; Takahashi et al., 2013). For instance, the works by Fujita and co-workers (Kikuchi et al., 1998; Fujita et al., 2000, 2002; Takahashi et al., 2013) with thin wires (~ 1 mm) revealed how the core size, oxygen concentration, and forced opposed flow can influence the flame spread rate

(FSR) along the wire insulation. Leung et al. (Leung et al., 2000) highlighted how the thermophysical properties and diameter of the core can accelerate or impede the pyrolysis and ignition of the insulation. Nakamura et al. (Nakamura et al., 2008, 2009) and Kobayashi et al. (Kobayashi et al., 2021, 2018, 2017) established that thermal conduction within the core plays a key role in determining the behavior of steady-state flame to spread, particularly when insulation thickness and gravity-driven dripping processes are considered.

A recurring theme in these studies is the dual role of the wire core on the flame spread characteristics. On the one hand, a highly conductive metal core (e.g., copper) can conduct heat ahead of the flame front, preheating the unburned insulation and enhancing the FSR. On the other hand, as shown by Kobayashi (Kobayashi et al., 2018) and Nakamura (Nakamura et al., 2008), the same conductive core can act as a heat sink if its diameter is large compared to the thickness of the insulation, or if it dissipates heat downstream of the insulation burning region, effectively cooling the insulation and altering the dynamics of heat transfer ahead of the flame or the melting and dripping of the insulation. This thermal balance strongly depends on the type and morphology of the wire conductor and insulation, the environmental conditions, and gravity. Together, these studies underscore both the complexity of the problem and the relevance of accurate predictive models. Despite the growing amount of experimental data and empirical correlations, the parameter space—which encompasses thermophysical properties, environmental conditions, and gravity value and orientation—is vast. As a result, traditional modeling approaches, whether simplified analytical models or CFD simulations, face challenges in covering the full range of scenarios reliably.

2. Motivation and problem definition

Although data-driven models have advanced our understanding of fire spread mechanisms and their sensitivity to material and environmental parameters, they are often constrained by strong assumptions and simplified boundary conditions. Furthermore, gathering the experimental data required to validate and calibrate such models under a broad range of conditions is costly and time-consuming. Advanced methods that can leverage existing datasets more efficiently and extrapolate beyond them are becoming increasingly helpful. This limitation is especially significant in spacecraft fire safety research and reduced gravity combustion, where flame spread experiments are costly, technically demanding, and inherently limited in number. Recent

reviews of reduced buoyancy and microgravity flame spread further emphasize the importance of improving predictive tools that can extract more value from sparse experimental datasets (Ma et al., 2025b).

Machine Learning (ML) techniques have emerged as powerful tools to handle high-dimensional data and discover complex, non-linear relationships among multiple parameters, especially useful in fire-related physical phenomena (Rivera et al., 2023). In the context of flame spread over electrical wires, there are only a few pioneering studies such as that of Gagnon et al. (Gagnon et al., 2021) that have explored preliminary approaches to predict flame spread over wires using artificial neural networks (ANNs). A recent related study by Ma et al. (Ma et al., 2025a) used a scale model framework combined with Bayesian Markov Chain Monte Carlo to predict the limiting oxygen concentration (LOC) of laboratory electrical wires under different gravity levels, while also systematically quantify parameter uncertainties. Although that work focused on LOC rather than flame spread rate, it further highlights the growing relevance of data-driven and reduced order predictive approaches for wire combustion problems. Although these studies show promising results, there remains an unexplored gap between the current Machine Learning predictive approaches and strategies employing deep learning, which warrants further investigation.

Machine Learning approaches can complement or even accelerate traditional modeling strategies, providing flexible predictive frameworks that integrate diverse input parameters such as core and insulation material properties, oxygen concentration, ambient pressure, forced flow velocities, and gravity. However, ML algorithms are typically “data-hungry” and may suffer from limited generalization if the training dataset does not encompass the full range of relevant conditions. This limitation is particularly important in fire-spread research, where the problem is complex (Fernandez-Pello, 1994) particularly in the case of electrical wires due to their morphological characteristics.

2.1. Problem definition and data limitation

Theoretical models for flame spread involve simplifying assumptions in order to produce analytical expressions for the FSR. An example is the analysis derived by Fernandez-Pello (Fernandez-Pello, 2003) for the oppose flow spread over thin solids, and shown in Equation 1,

$$V_f = l_h \left[\frac{\rho_s c_s \tau (T_p - T_o)}{\dot{q}_{fc}'' + \dot{q}_{fr}'' - \dot{q}_{rs}'' + \dot{q}_{ext}''} - C_2 t_{fl} \ln \left(1 - \frac{Da_{cr}}{Da_x} \right) \right]^{-1} \quad (1)$$

For instance, taking the heated length, l_h , as constant and neglecting complex kinetic effects, $t_{fl} \ln \left(1 - \frac{Da_{cr}}{Da_x} \right)$, reduces this equation to forms analogous to those developed by de Ris (De Ris, 1969) and Wichman et al. (Wichman et al., 1982), which are basically heat transfer models of the flame spread process. These models can be extended to wire flame spread by accounting for heat transfer within the wire core and its interaction with the insulation material. Even in these simplified analyses, parameters like core and insulation relative thickness and conductivity, gas properties, gravity, and external heat flux distributions must be known or estimated, making the prediction capabilities of the models challenging.

In practice, systematic measurements of all necessary parameters are not feasible, and the data reported in the literature are often non-uniform. This lack of comprehensive datasets hampers direct validation of theoretical models and limits their predictive capabilities. Thus, while existing correlations capture some essential physics of the problem, they fall short of incorporating the full range of parameters observed in experiments.

To address these challenges, this work introduces alternative methodologies that integrate ML and data augmentation techniques to enhance predictive modeling capabilities for flame spread rates over electrical wires. Ultimately, this study explores methods that guide future research towards more effective incorporating of physical principles into the machine learning prediction process during the training phase (Rivera et al., 2023). The contribution of this study lies not only in applying ML to a traditionally physics-driven domain, but also in introducing data augmentation methods—such as bootstrapping, noise injection, and the use of Generative Adversarial Networks (GANs). Although GANs are typically utilized for image-based data, here they are adapted and implemented to expand training datasets for the FSR problem artificially (Goodfellow et al., 2020; Creswell et al., 2018). By using data augmentation, the Machine Learning algorithms can better capture the underlying distributions and physical behaviors reflected in real experimental observations. This, in turn, can improve prediction accuracy, enhance model generalization, and potentially provide deeper insight into the relative importance of various input parameters.

A central emphasis of the present study is therefore the formulation of

a unified predictive framework for a literature derived wire-flame dataset that spans different materials, geometries, apparatuses, and operating conditions. Within that framework, categorical descriptors such as core material, insulation type, spread direction, and configuration are used to represent regime-dependent physical distinctions that are already known to influence flame spread behavior (Kobayashi et al., 2021, 2018; Nakamura et al., 2008, 2009). In this way, the use of categorization is not merely a numerical encoding step, but a structured means of organizing heterogeneous experimental observations so that machine learning models can better identify physically meaningful trends across otherwise fragmented datasets. This data centered formulation allows the study to evaluate how regime aware categorization and augmentation strategies influence attainable FSR prediction, while also establishing a more suitable basis for future incorporation of stronger physical constraints into data driven wire flame spread modeling.

3. Methodology

The methodology employed in this study integrates existing experimental data from literature sources to create a comprehensive dataset of flame spread behaviors over electrical wires. A Flow chart showing the methodology used is shown in Figure 1. The compiled raw dataset is first pre-processed and then used for systematic hyperparameter optimization of the selected machine learning algorithms through a grid search with 5-fold cross validation. The resulting best-performing hyperparameter configurations are subsequently used in the final predictive analysis, where model performance is evaluated through a 10-fold cross-validation across the different data arrangements considered in this study. The numbered branches in Figure 1 correspond to the five prediction outcomes analyzed later in the manuscript: (1) raw data without categorization, (2) categorized raw data (RD with categorization), (3) categorized raw data with bootstrap augmentation (RD+BT), (4) categorized raw data with noise injection (RD+NI), and (5) categorized raw data with GAN-based augmentation (RD+GANs).

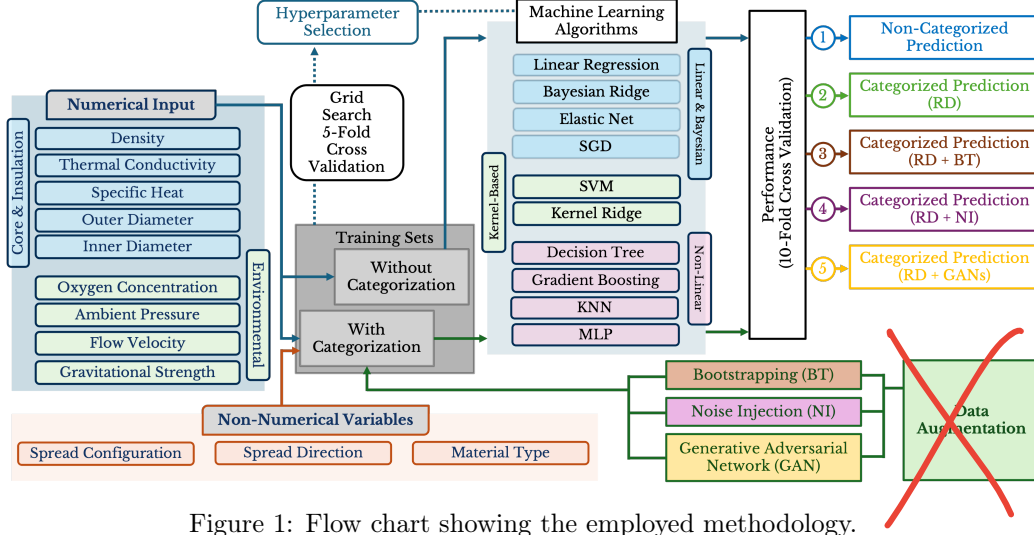


Figure 1: Flow chart showing the employed methodology.

The dataset considers **seventeen numerical input parameters**. Ten parameters pertain to the material and geometric properties of the core and insulation material: densities, thermal conductivities, specific heats, inner and outer diameters; and seven to the environmental conditions: oxygen concentration, ambient pressure, components of forced flow velocity vector (u_x , u_y , u_z) and gravity vector (g_x , g_y). **Sixteen additional non-numerical inputs** are considered to capture the spread configuration (vertical, horizontal, or inclined), spread direction (concurrent, opposed, or no flow), and the material type for core: Nichrome (NiCr), Copper (Cu), Steel, and Iron (Fe); and insulation: ethylene tetrafluoroethylene (ETFE), polyethylene (PE), low-density polyethylene (LDPE), high-density polyethylene (HDPE), and polymethyl methacrylate (PMMA). To make explicit the quantity and coverage of the compiled database, Table 1–2 summarize the numerical ranges and categorical options represented in the dataset. This summary also contextualizes the model generalization in sparsely sampled regions of the parameter space. Note that $\bar{v}$ and $\bar{g}$ summarize the range of imposed flow and gravity conditions across sources; in the model implementation, flow and gravity can also be represented via their components (u_x , u_y , u_z) and (g_x , g_y) when available.

The compiled database contains $N \approx 1200$ flame spread rate (FSR) data points extracted from published experiments spanning multiple wire geometries, material combinations, and environmental conditions. The raw dataset (RD) is used to establish the baseline model performance and to quantify the coverage of the input parameter space.

33 inputs

Table 1: Coverage of numerical predictors in the compiled flame spread dataset (raw data, RD; $N \approx 1200$). Reported values correspond to the global minimum and maximum across the compiled dataset.

Predictor	Units	Min	Max
Core density (ρ_c)	kg/m ³	7800 (NiCr)	8880 (Cu)
Core conductivity (k_c)	W/m·K	17.4 (steel)	398 (Cu)
Core specific heat ($c_{p,c}$)	J/kg·K	390 (Cu)	500 (steel)
Core diameter (d_c)	mm	0.30	5.5
Core cross-sectional area (A_c)	mm ²	0.071	24.0
Insulation density (ρ_i)	kg/m ³	923 (LDPE)	1700 (ETFE)
Insulation conductivity (k_i)	W/m·K	0.238 (ETFE)	0.338 (HDPE)
Insulation specific heat ($c_{p,i}$)	J/kg·K	1800 (ETFE)	2075 (HDPE)
Insulation thickness (τ)	mm	0.075	2.3
Insulation cross-sectional area (A_i)	mm ²	0.14	40.6
Oxygen concentration (X_{O_2})	vol%	14	50
Average pressure (P)	kPa	10	1000
Mean flow velocity ($\bar{v}$)	m/s	-3.5	6.8
Gravity level ($\bar{g}$)	g	10 ⁻⁵	1
Output: Flame spread rate, FSR	mm/s	0.0012315	15.0973

Table 2: Categorical predictors represented in the compiled dataset (RD).

Categorical predictor	Options represented
Core material	Copper, Steel tube, Iron, Nichrome
Insulation material	ETFE, PE, LDPE, HDPE, PMMA
Spread direction	Concurrent, Opposed, No Flow
Configuration/orientation	Horizontal, Vertical, Inclined

Ten different supervised Machine Learning Algorithms (MLA) are trained on the input–output relationships (Rivera et al., 2023), where the output is the flame spread rate along the wire (FSR). The MLA tested are as follows; Linear Regression (LR) (Goodfellow et al., 2016), Decision Tree (DT) (Breiman, 2017), K-Nearest Neighbors (KNN) (Goodfellow et al., 2016), Elastic Net (EN) (Goodfellow et al., 2016), Stochastic Gradient Descent (SGD) (Goodfellow et al., 2016), Bayesian Ridge (BR) (Goodfellow et al., 2016), Gradient Boosting (GB) (Friedman, 2001), Support Vector Machine

(SVM) (Smola and Schölkopf, 2004), Kernel Ridge (KR) and Multi-Layer Perceptron (MLP) (Goodfellow et al., 2016). Each of these algorithms produces a distinct model upon training, which encapsulates the learned parameters and extensive structures specific to the dataset.

3.1. Model training and hyperparameters

All models were implemented using scikit-learn and trained using a consistent preprocessing and evaluation pipeline to enable fair comparisons across model families. When categorical variables were included, they were incorporated via One-Hot encoding. Missing values were handled by imputing numerical features (mean fill as a first pass) followed by KNN imputation (KNNImputer with $n_{\text{neighbors}} = 1$), ensuring all algorithms were trained on complete feature matrices. Numerical features were standardized (StandardScaler) within each training fold and then applied to the corresponding validation data. To reduce the dependence of model comparisons on arbitrary default settings, systematic hyperparameter optimization was performed for each tunable algorithm using a grid search with 5-fold cross-validation (`GridSearchCV`) over a predefined search space appropriate to each model family. The selected MLAs span increasing flexibility and inductive bias: linear models (LR, EN, SGD, BR) provide interpretable baselines with varying regularization; tree and ensemble methods (DT, GB) capture nonlinear interactions and regime segmentation; KNN provides a local instance-based predictor; kernel methods (SVM, KR) provide nonlinear regression under smoothness assumptions; and MLP provides a flexible nonlinear model capable of learning higher-order feature interactions. Hyperparameter optimization was carried out using the original experimental dataset (RD), and the resulting best-performing configurations were subsequently applied to the categorized and augmented datasets. This strategy was adopted to preserve a consistent comparison framework across all data arrangements while reducing the risk that model rankings would be biased by unsuitable default parameter choices. The optimized hyperparameters selected for each model are summarized in Table 3. Model performance was then evaluated using random 10-fold cross-validation. This was adopted to provide a more robust estimate of predictive performance than a single train–test split while preserving sufficient training data within each fold. Because the compiled literature dataset is highly heterogeneous and unevenly distributed across sources, with some studies spanning multiple configurations and others representing only narrow experimental conditions, source-grouped validation was not adopted

here as the primary evaluation protocol (Leinonen et al., 2024). Accordingly, the reported cross-validation results should be interpreted as an assessment of model robustness within the compiled literature dataset, rather than as a strict leave-source-out test of transferability to a completely new and different experimental source.

Table 3: Best hyperparameter configurations selected by a grid search with 5-fold cross-validation on the original experimental dataset (RD).

Model	Selected hyperparameters
LR	No tunable hyperparameters (reference model)
DT	max_depth = 20, min_samples_split = 10, min_samples_leaf = 1
KNN	n_neighbors = 5, weights = distance
EN	alpha = 1, l1_ratio = 0.0
SGD	alpha = 0.1, penalty = elasticnet
BR	alpha_1 = 10^{-6} , lambda_1 = 10^{-4}
GB	learning_rate = 0.1, max_depth = 3, n_estimators = 200
SVM	kernel = rbf, C = 10, epsilon = 0.1
KR	kernel = rbf, alpha = 0.1, gamma = 0.01
MLP	hidden_layer_sizes = (100,), activation = relu, alpha = 0.01

3.2. Strategies

Initially, a baseline was established by training—testing the supervised MLA exclusively with numerical data (referred to as original or raw data, RD), while excluding classification inputs. In addition, we also investigate the effect of incorporating categorical variables, such as insulation material type and flame spread direction, since these non-numerical variables significantly influence the flame spread response (Kobayashi et al., 2021, 2018; Nakamura et al., 2008, 2009). Therefore, instead of using purely numerical features, a second baseline for predictions was created by incorporating categorical variables into the training process via One-Hot encoding. This approach transformed classification into binary indicators, allowing algorithms to leverage discrete categories more effectively and reveal hidden patterns.

Following this, three data augmentation strategies are introduced to improve model generalization, especially when new experimental data are scarce. Bootstrap resampling (BT); random resampling of data points from the existing dataset to produce new and varied training sets. Noise Injection (NI)

adding random perturbations to the training data to increase variability and improve model robustness. Generative Adversarial Network (GAN) using a neural network framework capable of synthesizing new, statistically similar data points. Synthetic data generated with GAN maintain similar distributions and physical patterns as the original data, thus extending the coverage of the dataset in a physically meaningful manner. By training on these augmented datasets, the ML models are expected to provide more robust and reliable FSR predictions.

3.3. Data augmentation implementation

To enable controlled comparisons among augmentation strategies, each method was used to generate an additional 1000 synthetic training points beyond the original RD dataset. Bootstrap resampling (BT) was implemented by sampling RD rows with replacement (random seed 42), increasing the density of empirically observed joint feature combinations without altering physical relationships. Noise injection (NI) was implemented by sampling 1000 RD rows and perturbing numerical predictors with additive Gaussian noise (mean 0, variance 10^{-6}) while retaining the corresponding categorical labels; this increases local variability while preserving the experimental regime classification. GAN-based augmentation was implemented using a fully connected generator–discriminator framework trained for 2000 epochs (batch size 32) with Adam optimizers (learning rate 10^{-4}); after training, 1000 synthetic samples were generated and used as an additional training set to expand coverage of the empirical feature distribution. Before GAN training, numerical features were normalized using a MinMaxScaler to map predictors to $[-1, 1]$, consistent with the activation of the generator output (tanh).

In this study, the GAN was trained on the joint distribution of the input features and target FSR under the working assumption that the experimental dataset embeds physically meaningful coupled trends among the variables. However, because the generator is purely data-driven, it does not explicitly enforce conservation laws, mechanistic constraints, or geometric consistency during training. Therefore, the GAN-generated samples should be interpreted as statistically consistent synthetic data rather than as strictly physics-constrained realizations. In the present wire-flame dataset, physical plausibility cannot be judged from a single variable alone, since FSR reflects the coupled influence of material properties, geometry, oxygen concentration, pressure, flow condition, and gravity (Konno et al., 2020; Kobayashi et al.,

2021). Bootstrap resampling naturally preserves the physical admissibility of the original observations, whereas noise injection and GAN-based augmentation may generate combinations that remain statistically plausible but are not guaranteed to satisfy all physical couplings. Accordingly, the role of GAN augmentation in this work is limited to expanding the empirical training distribution, while predictive performance is evaluated only through cross-validation on experimental data. These limitations further motivate the future development of physics-informed generative approaches for flame-spread datasets.

4. Results

The results of applying the above methodology to model flame spread over electrical wires using the different MLAs tested are presented here. Each algorithm yields a distinct fitted model whose internal parameters (e.g., tree splits, support vectors, or neural network weights) are specific to both the dataset and the selected hyperparameter configuration, and are therefore not directly interpretable in isolation. Accordingly, rather than listing fitted parameter values, we report (a) the model classes considered, (b) the training and hyperparameter optimization protocol described in Section 3.1, and (c) the predictive performance obtained under a consistent preprocessing and validation framework. Because the models were systematically tuned using cross validation, the reported results should be interpreted as comparative performance estimates for the evaluated dataset and data management strategies, rather than as baseline outcomes obtained from untuned default settings. Therefore, we summarize model performance using the standard regression metrics: (i) the coefficient of determination (R^2), which reflects how well the model explains the variance in the FSR, with higher R^2 values indicating better predictive capability; and (ii) the mean absolute error (MAE), which measures the average magnitude of the prediction errors. The performance values reported below correspond to tuned models evaluated using random 10-fold cross validation. A summary of the model performance metrics is presented in Table 4 and Table 5 for the different MLAs and data arrangements.

4.1. Impact of adding categorical variables

Examples of the impact of the use of categorical variables are presented in Figure 2, comparing the measured vs. predicted flame spread rates (FSR)

Table 4: Comparison of R^2 scores (mean $\pm$ std) across different datasets with and without categorization.

Model	No Categorization	With Categorization			
	RD	RD	RD + BT	RD + NI	RD + GANs
LR	0.53 ± 0.07	0.68 ± 0.04	0.68 ± 0.06	0.68 ± 0.05	0.65 ± 0.19
BR	0.51 ± 0.07	0.67 ± 0.05	0.67 ± 0.06	0.67 ± 0.05	0.65 ± 0.19
EN	0.39 ± 0.03	0.66 ± 0.04	0.66 ± 0.05	0.66 ± 0.04	0.67 ± 0.12
SGD	0.50 ± 0.06	0.65 ± 0.04	0.66 ± 0.05	0.65 ± 0.05	0.66 ± 0.13
KR	0.77 ± 0.05	0.89 ± 0.02	0.89 ± 0.03	0.88 ± 0.03	0.85 ± 0.05
SVM	0.81 ± 0.07	0.92 ± 0.02	0.93 ± 0.03	0.91 ± 0.03	0.84 ± 0.06
DT	0.80 ± 0.08	0.91 ± 0.03	0.91 ± 0.03	0.86 ± 0.04	0.77 ± 0.09
GB	0.86 ± 0.05	0.92 ± 0.02	0.95 ± 0.02	0.91 ± 0.02	0.88 ± 0.05
KNN	0.76 ± 0.07	0.91 ± 0.02	0.95 ± 0.03	0.90 ± 0.03	0.80 ± 0.07
MLP	0.80 ± 0.05	0.90 ± 0.02	0.91 ± 0.03	0.90 ± 0.03	0.80 ± 0.20

Table 5: Comparison of Mean Absolute Error (MAE) scores (mean $\pm$ std) across different datasets with and without categorization.

Model	No Categorization	With Categorization			
	RD	RD	RD + BT	RD + NI	RD + GANs
LR	1.04 ± 0.07	0.92 ± 0.09	0.93 ± 0.10	0.93 ± 0.08	0.68 ± 0.11
BR	1.12 ± 0.07	0.94 ± 0.09	0.97 ± 0.09	0.96 ± 0.07	0.68 ± 0.10
EN	1.25 ± 0.10	0.95 ± 0.09	0.96 ± 0.08	0.95 ± 0.07	0.67 ± 0.09
SGD	1.14 ± 0.08	0.95 ± 0.08	0.96 ± 0.07	0.96 ± 0.07	0.67 ± 0.09
KR	0.70 ± 0.05	0.44 ± 0.06	0.47 ± 0.06	0.46 ± 0.06	0.46 ± 0.06
SVM	0.49 ± 0.07	0.32 ± 0.05	0.30 ± 0.06	0.33 ± 0.05	0.43 ± 0.05
DT	0.50 ± 0.10	0.33 ± 0.06	0.32 ± 0.05	0.41 ± 0.05	0.52 ± 0.03
GB	0.46 ± 0.07	0.38 ± 0.07	0.35 ± 0.04	0.42 ± 0.04	0.41 ± 0.03
KNN	0.55 ± 0.09	0.34 ± 0.07	0.19 ± 0.04	0.36 ± 0.06	0.50 ± 0.06
MLP	0.56 ± 0.07	0.38 ± 0.06	0.39 ± 0.06	0.40 ± 0.06	0.46 ± 0.08

using a linear model (Bayesian Ridge) and a nonlinear model (Multi Layer Perceptron). Column (a) presents the results for non-categorized data and (b) shows the results of training the same algorithms but using categorized data.

When algorithms receive only numerical parameters (no categorization) such as diameters, pressures, or velocities, simpler methods such as LR or BR produce low R^2 (~ 0.5), indicating difficulty in capturing flame spread (FS) interactions, as shown in Figure 2a. More flexible, tree-based, and ensemble models (e.g., Decision Trees, Gradient Boosting) partially succeed ($R^2 \sim 0.8$) because they can isolate certain non-linearities. However, neither approach explicitly recognizes different flame-spread regimes when no categorical input is provided.

On the other hand, when training the MLAs with categorization (RD only), each experiment is labeled according to its specific FS regime—like core material type—all models benefit because the physics of conduction,

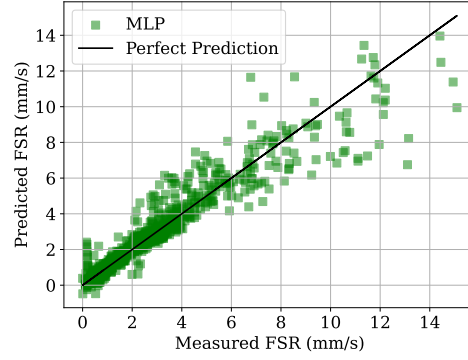
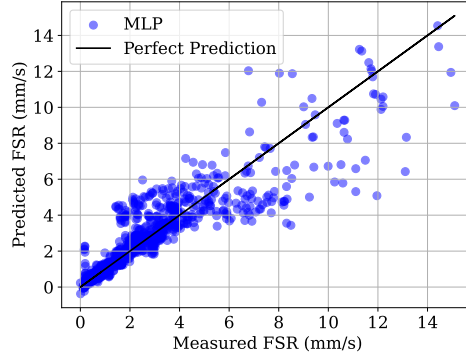
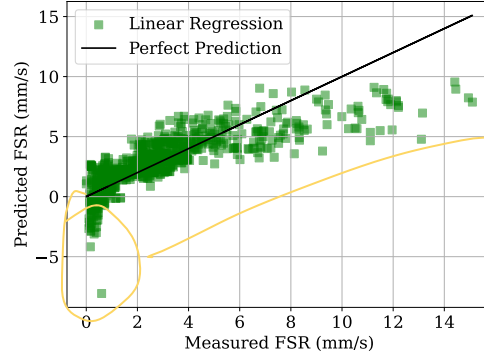
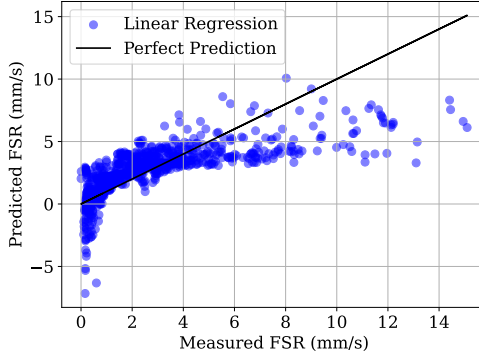
BR
 $R^2 \sim .5$



MLP
 $R^2 \sim .8$



NON
CONSTRAINED



(a) Without categorization.

(b) With categorization.

Figure 2: Predicted vs. Measured FSR and perfect prediction line (PP) for linear and non-linear models.

convection, or dripping is more readily distinguished. Even linear methods (LR, EN, BR) show better performance ($R^2 \geq 0.68$). This means that they can differentiate between phenomena such as heat-sink effects in high-conductivity core materials under (Kobayashi et al., 2021, 2018; Nakamura et al., 2008, 2009; Leung et al., 2000; Kikuchi et al., 1998; Zhao et al., 2017; Guo et al., 2024; Fujita et al., 2000, 2002; Takahashi et al., 2013; Kobayashi et al., 2017). Meanwhile, non-linear algorithms (Decision Trees, KNN, MLP, Gradient Boosting) exceed $R^2 \sim 0.88$ because they naturally segment data based on these regime-defining flags, aligning well with the real-world diversity of FS behaviors. It should also be noted that a small number of predictions in Figures 2 and 3 fall below zero. Since the flame spread rate (FSR) is physically nonnegative, these values are not interpreted as admissible physical predictions. Rather, they arise because the regression models

used here are unconstrained real-valued estimators and therefore do not enforce a zero lower bound on the output. This effect appears primarily in the low FSR regime, where the spread rate approaches its lower limit and even modest residual errors, combined with the natural variability of the dataset, can cross below zero. This behavior indicates reduced model reliability near very low spread rates and motivates future use of constrained-output, transformed-target, or physics-informed regression strategies.

The observed differences align with the non-linearity of flame spread behavior. Factors such as insulation material type, core conductivity, and spread direction define different heat transfer and combustion regimes. Categorization made these regimes more clear, allowing models to identify scenario-dependent patterns. Therefore, the improvements in R^2 and MAE indicate that the models are better able to isolate and leverage these physically significant differences. The reduced MAE, in particular, means that models are consistently closer to the actual FSR values, not just on average but also across the spectrum of conditions. This suggests that models are learning relationships aligned with actual heat transfer and combustion physics—e.g., how adjusting oxygen concentration or changing insulation material fundamentally alters the flame spread process.

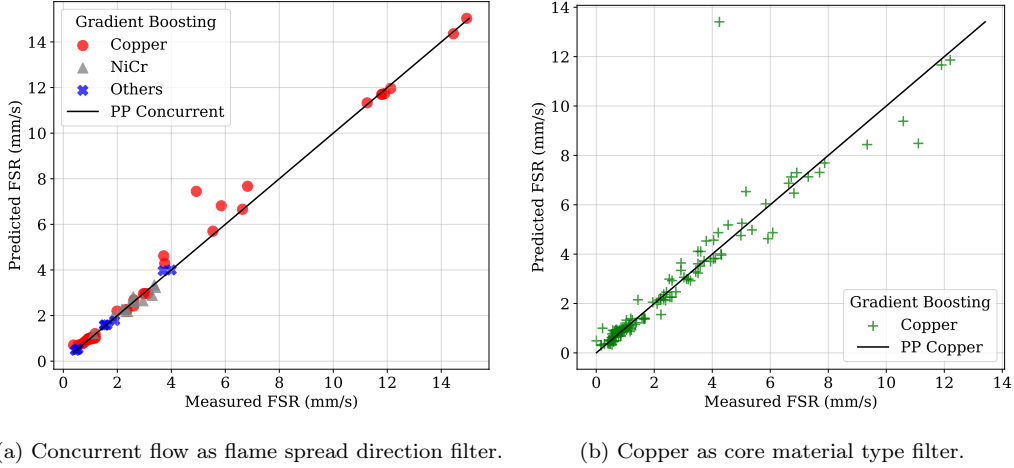


Figure 3: Predicted vs. Measured FSR and perfect prediction line (PP) using categorization strategy. These figures visualize predictions across the entire dataset using the best-performing model to demonstrate the performance for specific configurations.

Although we are not directly using physics constraints as Physics In-

formed Machine Learning (PIMLs) would, the use of non-numerical features remains a rudimentary step toward embedding physical logic. For instance, defining the flame spread direction relative to the gas flow direction as “concurrent flow” can stabilize the flame front, producing more uniform convective heating with less turbulence and a more straightforward balance of energy, leading to a high R^2 value of 0.98 in our model’s predictions (Figure 3a). Here, the primary controlling parameters (core conductivity, insulation thickness, local oxygen concentration) interact in a more predictable way when the flame moves in the same direction as the forced flow. Under concurrent flow, gas phase heating of the virgin insulation can become more dominant, while the relative influence of solid phase axial heat conduction and other competing heat loss pathways is reduced. As a result, the mapping between the input variables and the resulting flame spread rate becomes more regular, which is easier for the model to learn. This interpretation is consistent with the analysis reported by Ma et al. (Ma et al., 2022), who showed that the controlling heat transfer behavior over the preheating zone differs fundamentally between upward/concurrent-like and downward/opposed-like flame spread over electrical wires. If the focus goes to “copper” as the core material category, it also highlights the role of thermal conductivity in driving pyrolysis and flame spread. Restricting the dataset to copper yielded $R^2 = 0.89$, higher than the non-categorized scenario. Copper has a high thermal conductivity and distinctive heat conduction pathways within the insulation layer (importance of geometric factors). This property can lead to a clearer, more consistent trend in how heat is transferred to the insulation and subsequently affects pyrolysis and flame spread. Thus, the data-driven approach suggests that a more robust theoretical model of flame spread rate (FSR) might benefit from considering piecewise behaviors and regime-specific correlations, rather than being condensed into a single formula without recognizing regime changes.

4.2. Impact of data augmentation techniques

After establishing that categorization generally improved model performance, the next step was to evaluate the influence of various augmentation strategies—BT, NI, and GAN—on the predictive capabilities of the tuned ML models. This evaluation involved re-training the selected MLAs on newly augmented training datasets. Each dataset was modified according to one of the augmentation strategies, allowing us to directly compare how these methods affect model accuracy and overall effectiveness in predicting the outcomes. Figure 4 shows how the implemented strategies generate syn-

thetic data compared to the original distribution. This logic applies to all inputs, though only the results for the FSR are displayed here.

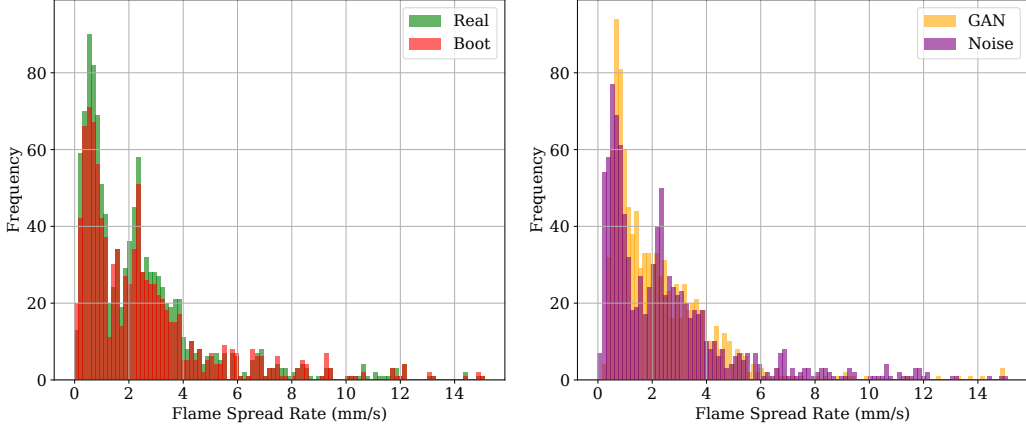


Figure 4: Histograms of FSR with the proposed data augmentation strategies.

Although all three strategies generally follow the observed flame spread rate (FSR) distribution, Bootstrap sampling (BT) performs particularly well in the low-FSR region, an area with higher data density. BT retains the original physical relationships, but resamples them more thoroughly. Among the tested strategies, BT provides the most consistent gains for tree-based and neighborhood-based methods (e.g. Decision Tree, Gradient Boosting, KNN), with mean R^2 values often exceeding 0.90. GAN-generated data, showing reasonable agreement with the empirical distribution, attempt to synthesize new yet statistically patterned samples within the feature space. In the tuned-model setting, GAN augmentation remains beneficial for some simpler algorithms such as LR and BR, but its effect is more mixed overall and does not systematically outperform categorization alone or bootstrap augmentation. This suggests that, although GAN-based augmentation can help expand the empirical parameter space, its benefits depend on how well the synthesized samples preserve the coupled structure of the original data. Certain methods, such as SVM, can still degrade under GAN if synthesized samples are less compatible with the kernel representation selected for the algorithm. Noise Injection shows some improvement—particularly for simpler linear-type models—but rarely matches the impact of BT. The reason might be that NI can create parameter perturbations that are not entirely

physically consistent—e.g., adjusting material density without corresponding changes in other dependent properties—thus generating outliers that do not reflect realistic flame-spread scenarios. Although NI can highlight small-scale sensitivities, this benefit must be weighed against the risk of introducing implausible data points.

Bootstrap resampling preserves physical consistency by construction because it resamples existing observations from the experimental dataset. In contrast, noise injection perturbs numerical predictors locally around experimental points while retaining their categorical labels, and GAN-based augmentation learns the joint statistical distribution of the data; therefore, neither NI nor GAN explicitly guarantees algebraic consistency among derived geometric quantities (e.g., consistency between diameter and cross-sectional area) or exclusion of all physically impossible material-property combinations. For this reason, NI- and GAN-generated samples are interpreted here as statistically plausible synthetic data rather than as strictly physics constrained realizations, which motivates future development of constrained or physics informed generative augmentation strategies.

4.3. Comparison of model families and their response to augmentation

When hyperparameters are systematically tuned, the relative behavior of the different model families becomes clearer. Linear and Bayesian methods (LR, BR, EN, SGD) remain more limited than the best-performing nonlinear approaches, but they still benefit substantially from physically meaningful data structuring. In particular, categorization consistently improves these models, raising their mean R^2 values into the range of approximately 0.65–0.68 across most categorized cases. This indicates that, once the regime structure is made more explicit, even simpler models can capture a meaningful portion of the dominant dependencies linking material properties, geometry, oxygen concentration, pressure, and flow conditions to FSR. Kernel-based methods (SVM, KR) are, in principle, well suited to capture nonlinearities in FSR, but their effectiveness depends strongly on kernel choice and associated hyperparameters. In the present study, systematic tuning yielded substantial improvements for these models, showing that their predictive capability is highly sensitive to how the input space is represented. For example, KR is highly sensitive to kernel specification; in the present case, selecting a nonlinear RBF kernel during tuning substantially improved its ability to represent the coupled structure of the FSR dataset. This point is particularly important because, if KR is evaluated with a linear kernel, its performance

becomes poor and can even yield negative R^2 values, indicating that a linear representation is inadequate for the present flame spread problem. By contrast, with an appropriate nonlinear kernel and tuned regularization, KR becomes competitive, while SVM reaches mean R^2 values above 0.9 in the categorized datasets. These results indicate that kernel methods can provide reliable predictive performance when both the feature representation and the hyperparameter selection are appropriate for the coupled, non-monotonic relationships governing FSR. Nonlinear models (DT, GB, KNN, MLP), by design, map nonlinear interactions more effectively and remain among the strongest performers overall. Tree-based and neighborhood-based methods, particularly GB and KNN, stand out when combined with categorization and bootstrap augmentation, achieving the highest and most consistent R^2 values with relatively low variability. This suggests that they can effectively separate and model distinct flame spread regimes (e.g., horizontal vs. vertical, high vs. low-pressure, or different material combinations) without oversimplifying the underlying trends. MLP also performs strongly after tuning, especially when categorization is included, although its gains are somewhat less pronounced than those of GB and KNN under bootstrap augmentation. Performance of selected nonlinear models is presented in Figure 5. Because the compiled dataset is strongly concentrated in the low-FSR regime (see Figure 4), parity plots covering the full response range can visually obscure differences in model behavior at higher spread rates. Most available data fall below approximately 4 mm/s, whereas the 6–16 mm/s region is comparatively sparse. This imbalance contributes to the larger scatter observed at high predicted values, where fewer training examples are available and the fitted response is less constrained. To improve interpretability, Figure 5 shows the two nonlinear models that best represent the improvement relative to the RD baseline (KNN and GB), and each case is displayed over separate FSR intervals: 0–6 mm/s on the left and 6–16 mm/s on the right. A representative sampling of points is shown in the dense low-FSR region for visual clarity, while the sparse high-FSR region is displayed in full.

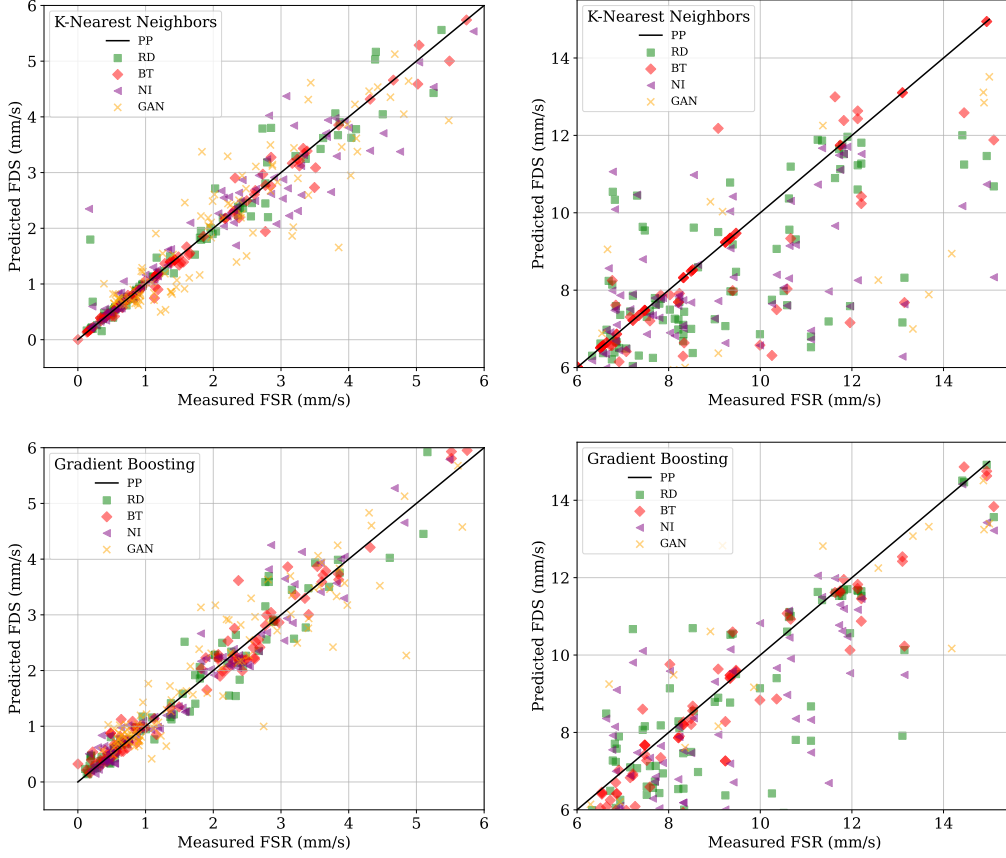


Figure 5: Performance of K-Nearest Neighbors (top) and Gradient Boosting (bottom) models. Parity plots are separated into two FSR intervals: 0–6 mm/s (left) and 6–16 mm/s (right). A representative sampling of points is shown in the low-FSR interval for clarity.

Figure 6 summarizes the mean R^2 values obtained for the tested ML models across the different data-management strategies, grouped by model family. Each marker represents one ML algorithm and each color corresponds to a strategy: blue for RD without categorization, green for RD with categorization, red for RD+BT, purple for RD+NI, and orange for RD+GANs. Two trends become evident. First, categorization consistently improves performance across nearly all model families, confirming that explicit regime information is highly beneficial in this problem. Second, bootstrap augmentation remains the most robust augmentation strategy, particularly for GB and KNN, and is the configuration most consistently associated with the

highest mean R^2 values and relatively low dispersion. By contrast, GAN-based augmentation produces a more mixed response: it remains competitive in some cases, but does not systematically outperform categorization alone and is often associated with larger variability. Overall, the tuned results show that the strongest predictive performance is obtained when physically meaningful categorization is combined with a stable augmentation strategy such as bootstrap resampling.

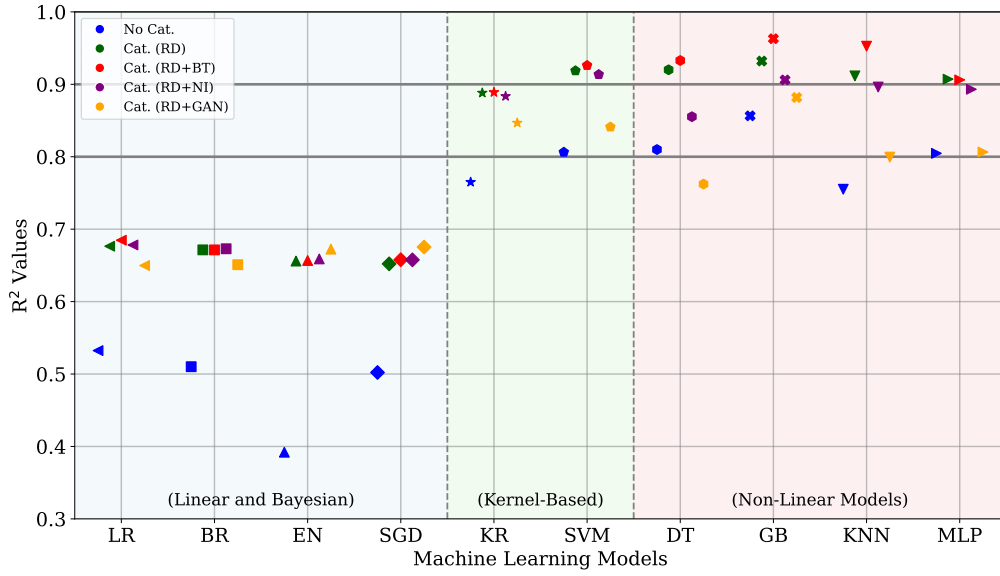


Figure 6: Summary of mean R^2 results for the tested ML models and data-management strategies.

5. Conclusions

This study found that integrating categorical variables and data augmentation techniques into data-driven models can significantly enhance the predictive accuracy and interpretability of flame spread rate (FSR) predictions on electrical wires. By incorporating categorical inputs that reflect distinct physical regimes, such as insulation material type, core properties, and spread direction, this work has effectively introduced a form of domain knowledge into otherwise purely numerical datasets. The results show that algorithms, especially non-linear approaches like decision trees and gradient boosting, benefit greatly from these structured hints, yielding higher R^2

values and lower errors (MAE). The improved stability and accuracy across several conditions indicate that acknowledging regime-specific behaviors is important to refine predictive capabilities.

Beyond categorization, data augmentation strategies—namely Bootstrap sampling, Noise Injection, and Generative Adversarial Networks (GANs)—help mitigate the sparse and heterogeneous nature of experimental flame spread data. Bootstrap sampling proved especially beneficial for tree based and neighborhood based models, elevating R^2 above 0.90 in many cases by comprehensively resampling existing physical scenarios. Meanwhile, GAN generated synthetic data offered support to simpler algorithms like Linear Regression and Bayesian Ridge, suggesting that filling gaps in parameter space enables these methods to approximate key heat-transfer relationships more effectively. However, kernel based methods occasionally suffered when augmented samples deviated from their underlying distribution assumptions.

Taken together, these findings highlight that bridging physical knowledge (e.g. material dependent conduction, flow-oriented heat transfer) and data driven modeling (including categorization and strategic augmentation) can significantly improve FSR predictions. Nevertheless, the current theoretical framework remains incomplete, limiting the direct application of fully physics informed models such as PINNs or PIMLs. This study’s insights, particularly those gained through classification and segmentation of FSR data according to physically meaningful conditions, help chart a path toward the refinement of the theoretical framework. By better understanding how different regimes interact and which parameters are most influential, future modeling efforts can more effectively incorporate theoretical constraints.

Acknowledgements

The work presented here was supported by NASA Grants 80NSSC17M0065 and 80NSSC22K0582.

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